

A Linear and Scalable Cutting-Plane Algorithm for Electricity Pricing

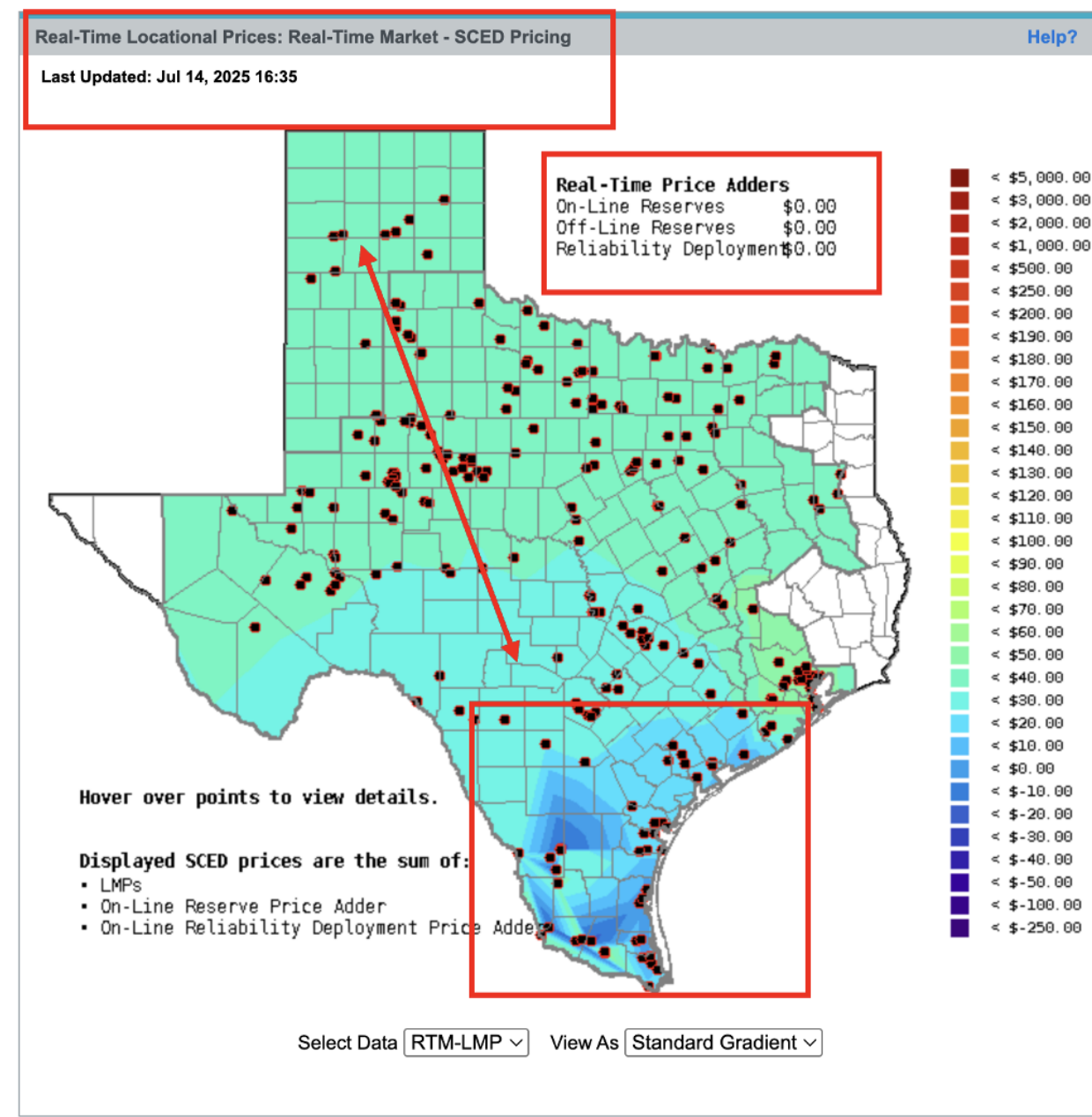
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Motivation and Goals



- Electricity pricing is highly complex:
 - Updates are made close to real-time (5 min. interval).
 - Grid reliability must be accounted for at all times.
 - Network congestion can create large spatial gradients in nodal prices.
 - Changing conditions in demand and supply can significantly affect price volatility.
- Existing market-clearing models often use DC approximations, which are computationally efficient but ignore key non-convexities.
- Price signals can be very uninformative in particular in congested networks.

Goals

- Develop a pricing algorithm that achieves computational performance comparable to DC-OPF, but with improved modeling accuracy of AC grid behavior.
- Evaluate our algorithm under two established pricing rules and observe strong empirical performance on the largest ARPA-E GOC2 instances.

ACOPF – Welfare Problem

$$(P) \quad \max \sum_{l \in \mathcal{L}} \pi_l(p_l) + \sum_{g \in \mathcal{G}} \pi_g(p_g, x_g)$$

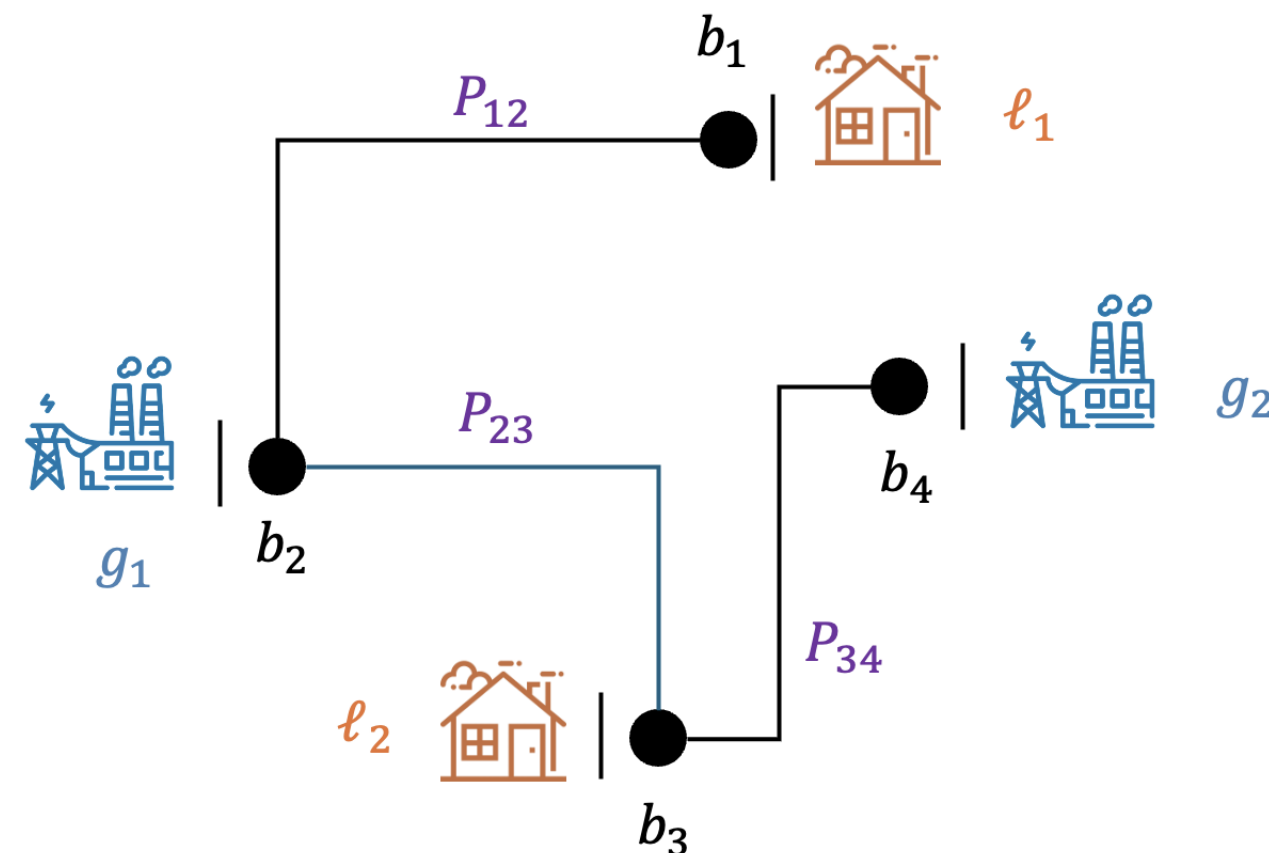
$$\text{s.t. } (p_l, p_g)_{l \in \mathcal{L}, g \in \mathcal{G}} \in \text{Bal}(\mathcal{N}),$$

$$(p_l, p_g)_{l \in \mathcal{L}, g \in \mathcal{G}} \in \text{Flow}(\mathcal{N}),$$

$$p_l \in \mathcal{P}_l, \quad \forall l \in \mathcal{L}$$

$$p_g \in \mathcal{P}_g(x_g), \quad \forall g \in \mathcal{G}$$

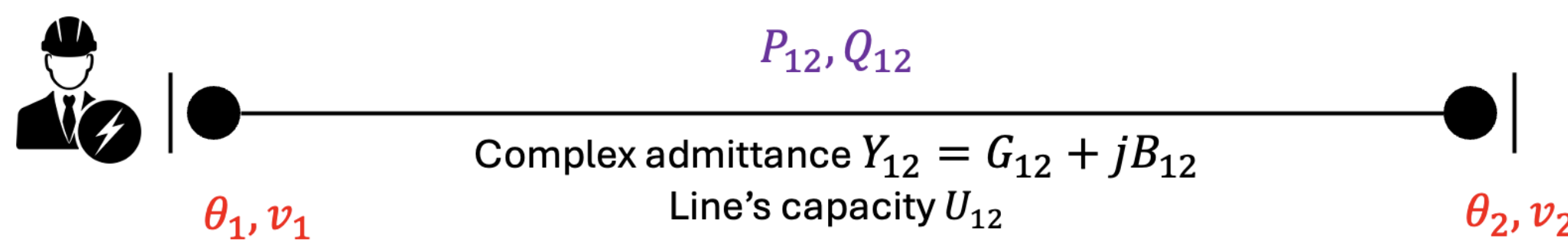
$$x_g \in \{0, 1\}^{\mathcal{G}}, \quad \forall g \in \mathcal{G}$$



- Active power balance: $\sum_{e \in \delta(b)} P_e = \sum_{g \in \mathcal{G}_b} p_g - \sum_{l \in \mathcal{L}_b} p_l$,
- Reactive power balance: $\sum_{e \in \delta(b)} Q_e = \sum_{g \in \mathcal{G}_b} q_g - \sum_{l \in \mathcal{L}_b} q_l$.

Power Flow Constraints

Operator controls voltage levels v and angles θ , and must keep active and reactive power balance in the network.



Alternating Current power flow definition:

$$P_{12} = G_{11}v_1^2 + G_{12}v_1v_2 \cos(\theta_1 - \theta_2) + B_{12}v_1v_2 \sin(\theta_1 - \theta_2)$$

$$Q_{12} = -B_{11}v_1^2 + B_{12}v_1v_2 \cos(\theta_1 - \theta_2) - G_{12}v_1v_2 \sin(\theta_1 - \theta_2)$$

$$P_{12}^2 + Q_{12}^2 \leq U_{12}^2$$

Direct Current approximation:

$$P_{12} = B_{12}(\theta_1 - \theta_2)$$

$$Q_{12} = 0$$

$$P_{12} \leq U_{12}$$

Jabr Second Order Cone relaxation [1]:

$$P_{12} = G_{11}v_1^{(2)} + G_{12}c_{12} + B_{12}s_{12}$$

$$Q_{12} = -B_{11}v_1^{(2)} + B_{12}c_{12} - G_{12}s_{12}$$

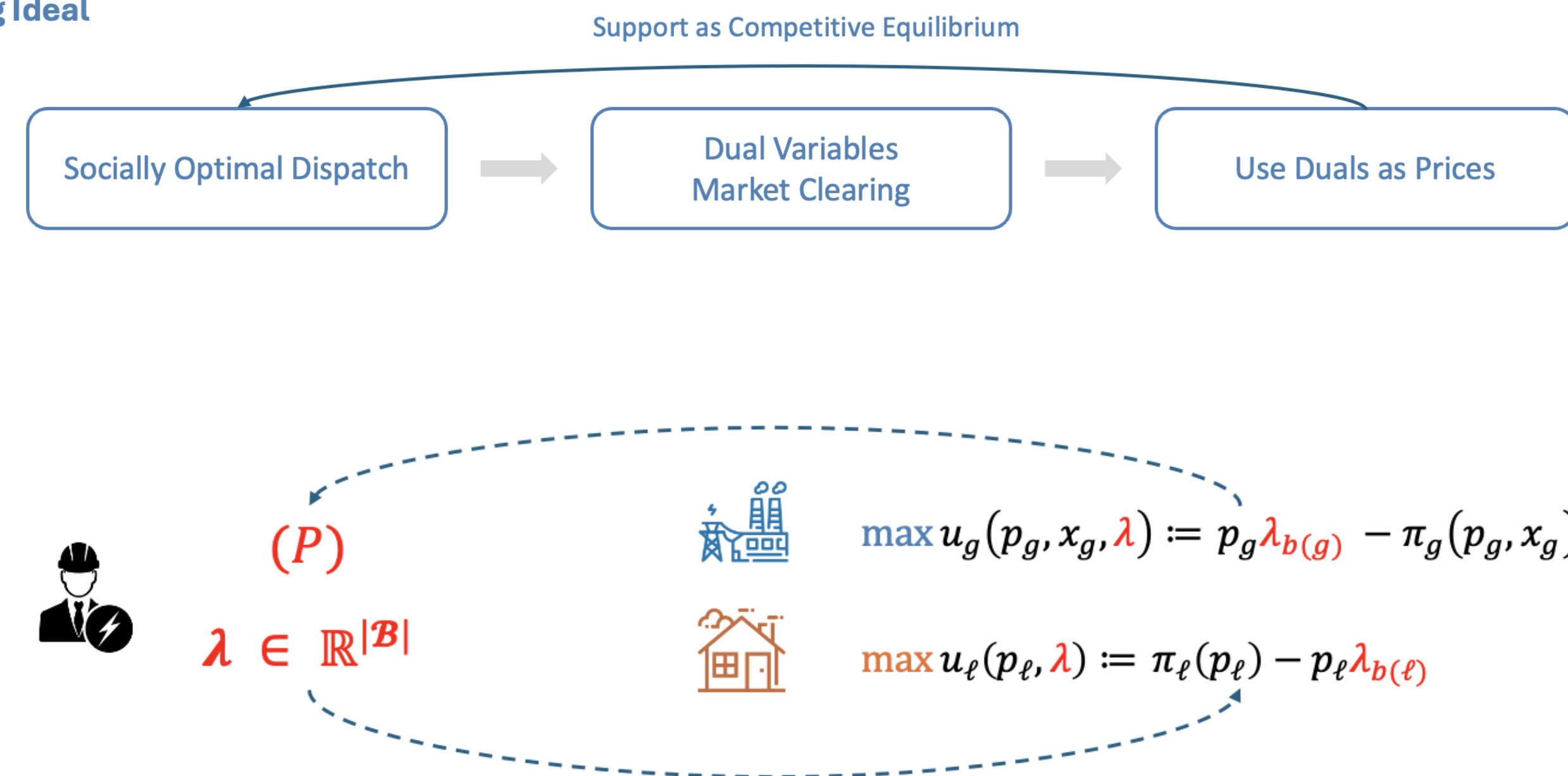
$$P_{12}^2 + Q_{12}^2 \leq U_{12}^2$$

$$c_{12}^2 + s_{12}^2 \leq v_1^{(2)}v_2^{(2)}$$

Pricing

System operator wants to find a price vector $\lambda \in \mathbb{R}^{\mathcal{B}}$ such that the individual decisions of utility-maximizing agents $\mathcal{G}, \mathcal{L}$ correspond to an allocation which is feasible and socially optimal.

Pricing Ideal



A price vector is **efficient** if the solutions to $\arg \max u_g(p_g, x_g, \lambda)$ and $\arg \max u_\ell(p_\ell, \lambda)$ solve (P) .

Theorem 1 (Sarf, 1954)

An efficient price vector is not guaranteed to exist under non-convex payoffs or feasible regions.

Let $z = (p, x)$ correspond to the allocation obtained by the planner. Dispatch must belong to the feasible region. Denote by $\phi(z)$ an AC-feasible solution obtained from z .

$$\forall \tau \in \mathcal{L} \cup \mathcal{G} : \sup_{z'} \{u_\tau(z', \lambda)\} - u_\tau(\phi(z), \lambda)$$

LOC or Incentive Compatibility

$$\forall \tau \in \mathcal{L} \cup \mathcal{G} : [-u_\tau(\phi(z), \lambda)]^+$$

MWP or Individual Rationality

$$\forall \tau \in \mathcal{L} \cup \mathcal{G} : [u_\tau(z, \lambda) - u_\tau(\phi(z), \lambda)]^+$$

RDC or Network Feasibility

Desirable Properties for an Electricity Pricing Algorithm

- λ satisfies linearity and homogeneity, i.e., it is non-discriminatory.
- λ maintains low MWP, LOC, RDC.
- λ can be computed in real time.

State of the Art

Pricing Rules. Electricity pricing literature focuses mainly on non-convexities x_g via Integer Programming Pricing [4] and Convex Hull Pricing [5]. Can incur in **side-payments** or **complex computations**.

Algorithm 1 Integer Programming (IP) Pricing

- $p_g^*, x_g^*, p_l^* \leftarrow \text{Sol}(P)$
- $\tilde{P} \leftarrow P + \{x_g = x_g^* \forall g \in \mathcal{G}\}$
- $\tilde{D} \leftarrow \text{Dual}(\tilde{P})$
- $\lambda^{\text{IP}} \leftarrow \text{Bal}(\mathcal{N}) \subseteq \text{Sol}(\tilde{D})$

Algorithm 2 Convex Hull (CH) Pricing

- $\tilde{P} \leftarrow \text{Conv}(P)$
- $\tilde{D} \leftarrow \text{Dual}(\tilde{P})$
- $\lambda^{\text{CH}} \leftarrow \text{Bal}(\mathcal{N}) \subseteq \text{Sol}(\tilde{D})$

Non-Linear Flows. Recent work evaluates pricing under convex flow relaxations in small networks [2]. Second-order cones and SDP provides better price signals but computation may be prohibitive.

We propose an iterative cutting-plane pricing algorithm (CPPA) to overcome this difficulty.

Table 1. Benchmark Flow Models

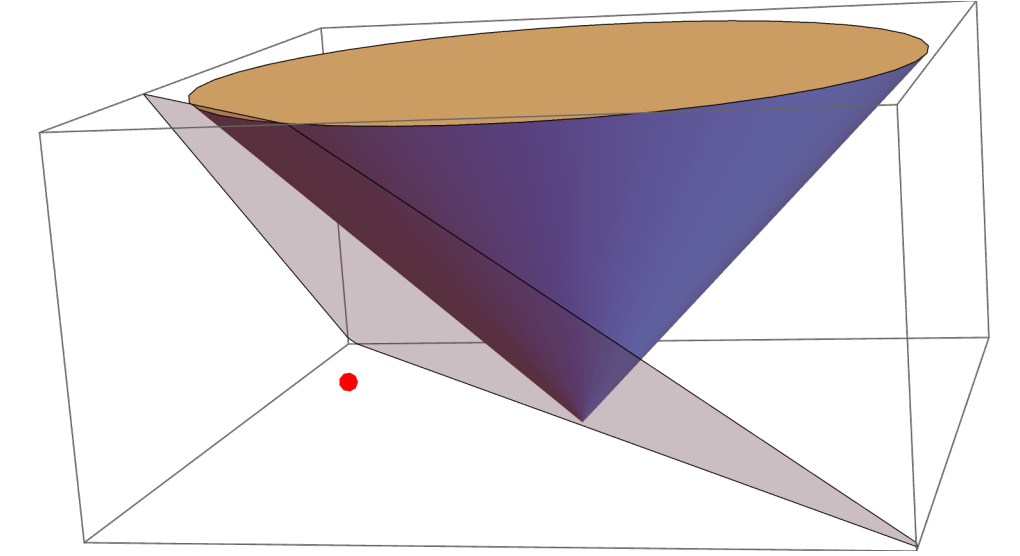
Power Flow	DC	SOC	AC	CPPA
Class	Linear	Non-Linear	Non-Linear	Linear
Feasible Region	Convex	Convex	Non-Convex	Convex
AC Region	Approximation	Relaxation	True Region	Relaxation

The Algorithm

$M_0 \leftarrow$ SOC Relaxation with **only linear constraints**

Algorithm 3 Cutting-Plane Pricing Algorithm

- Initialize $M \leftarrow M_0$
- If CH Pricing **Then** Relax x_g
- while** $t < T$ **do**
- $\bar{x} \leftarrow \arg \min M$
- Compute **cuts** for top $p_i\%$ violated branches
- Add cuts to M if they are not p_i -parallel to cuts in M
- Drop** cuts of age $\geq T_{age}$ whose **slack** is ϵ_i
- end while**
- Solve M
- If** IP Pricing **Then** Fix x_g in M and re-solve
- Extract dual variables λ of $\text{Bal}(\mathcal{N})$ as prices



$$c_{12}^2 + s_{12}^2 \leq v_1^{(2)}v_2^{(2)}$$

$$\iff \left\| \begin{pmatrix} 2c_{12} \\ 2s_{12} \\ v_1^{(2)} - v_2^{(2)} \end{pmatrix} \right\|_2 \leq w + z$$

- Cuts are **sparse** and **cheap** to compute: $(\bar{x}, \bar{t}) \notin \{(x, t) \in \mathbb{R}^n \times \mathbb{R}_+ : \|x\|_2 \leq t\}$ then

Generic outer-envelope cut: $\bar{x}^\top x \leq \|\bar{x}\|_2 t$

Jabr cut branch (1, 2) $\alpha_{12}c_{12} + \beta_{12}s_{12} + \gamma_{12}v_1^{(2)} + \zeta_{12}v_2^{(2)} \leq 0$

- Warm-starts:** “We don’t start from scratch” Our cuts are structural!

Numerical Results

Table 2. Economic Efficiency I (617 Avg.)

Model	Welfare	RDC
Jabr	654,730	330
CPPA	654,720	710
DC	653,630	6,710

Table 3. Economic Efficiency II (617 Avg.)

Model	MWP	LOC
Jabr	32,390	194,590
CPPA	32,390	194,610
DC	32,310	196,030

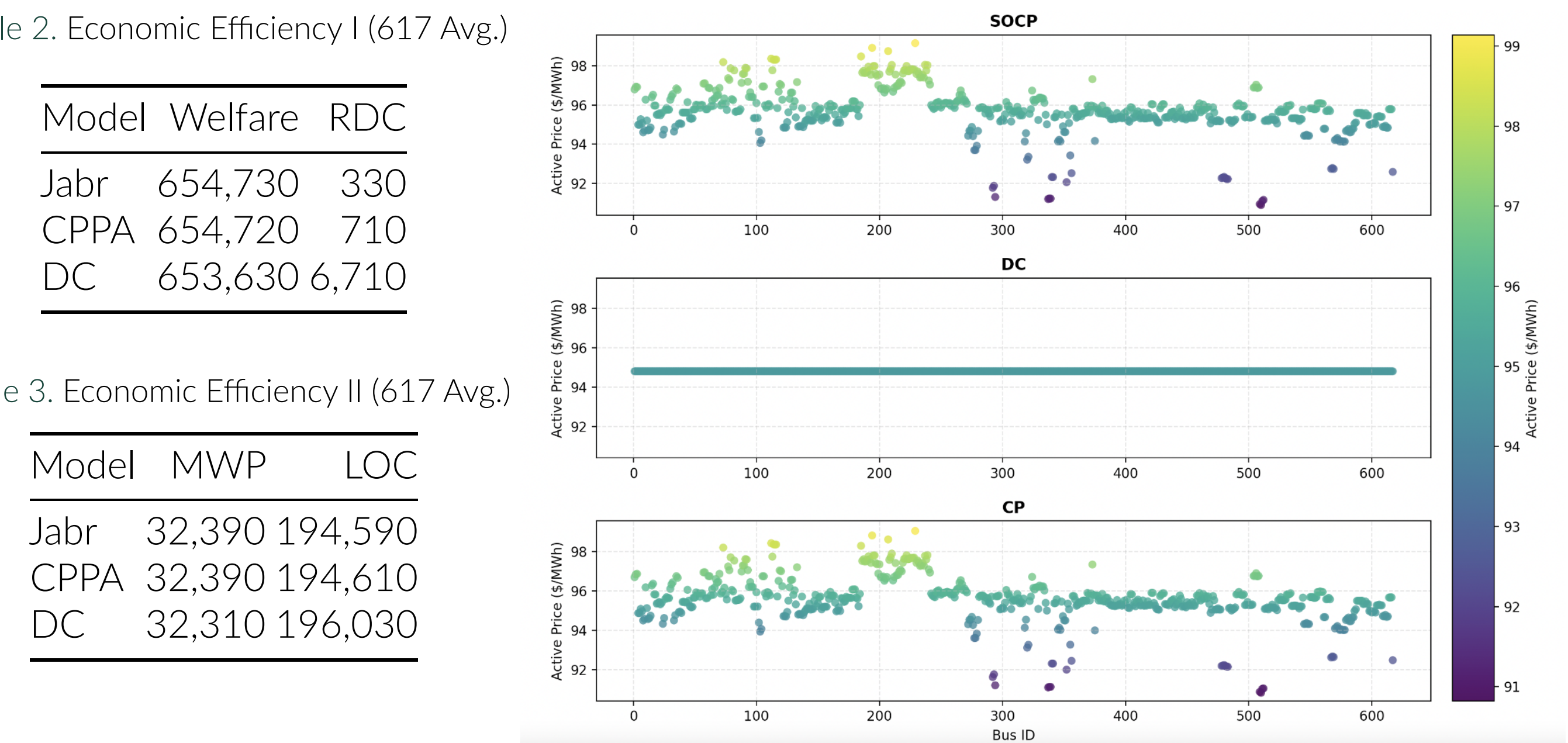


Table 4. Scalability (31,777 Avg.)

Model	Avg	Std	Time	δ	%Gap
Jabr	3.39	0.84	139.43	-	-
CPPA	3.37	0.58	42.16	0.03	0.00
DC	3.96	10.27	36.60	1.23	0.14

(*) Time Cut: 279.50

Table 5. Warm-Starts (31,777)

Model	Case 015			Case 011		
	Avg	Std	Time	Avg	Std	Time
Jabr	3.45	3.66	149.27	*	-	162.63
CPPA	3.42	3.60	72.26	Infeas	-	78.97
DC	4.33	12.20	36.23	5.09	13.14	36.47

Selected References

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